Module 6a slides: Multiple Oral Doses (worked examples)

Module 6 - Multiple Dosings · Lecture slides, worked examples · 28 Sep 2026

Name _____ Date _____

Example 1 An adult male (75 kg) received 250 mg tetracycline hydrochloride orally every 8 hours for 2 weeks. Tetracycline hydrochloride is about 75% bioavailable and has a V_D of 1.5 L/kg. The t½ is about 10 hours and the k_a is 0.9 hr^{-1}. When does the peak of the first dose occur?

Example 1 For the tetracycline regimen (250 mg orally, F 0.75, V_D 112.5 L, k 0.0693 hr^{-1}, k_a 0.9 hr^{-1}), the first dose peaks at 3.09 hours. What is C_{max} of the first dose?

Example 1 For the tetracycline regimen (250 mg orally every 8 hours, k 0.0693 hr^{-1}, k_a 0.9 hr^{-1}), when does the peak occur at steady state?

Example 1 For the tetracycline regimen (250 mg orally every 8 hours, F 0.75, V_D 112.5 L, k 0.0693 hr^{-1}), the steady-state peak occurs at 2.06 hours. What is C_{max} at steady state?

Example 1 For the tetracycline regimen (250 mg orally every 8 hours, F 0.75, V_D 112.5 L, k 0.0693 hr^{-1}, k_a 0.9 hr^{-1}), what is C_{min} at steady state?

Example 1 For the tetracycline regimen (250 mg orally every 8 hours, F 0.75, V_D 112.5 L, k 0.0693 hr^{-1}), what is Cavg at steady state?